An Explainable Attention-Enhanced Deep Learning Framework for Healthy and Bleached Coral Classification

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***Abstract*— Coral bleaching is a major threat to reef ecosystems, creating a need for accurate and scalable methods for automated coral-health assessment. This study proposes CoralNet, a robust deep learning framework for binary classification of healthy and bleached corals from underwater images. Three publicly available coral-image datasets were integrated to construct a unified dataset containing 11,656 images, comprising 5,632 healthy and 6,024 bleached coral samples. CoralNet employs ConvNeXt-Tiny as its primary convolutional backbone and incorporates attention-based and multi-scale feature representation mechanisms to improve the extraction of discriminative coral features. The training framework integrates AdamW optimization, cosine learning-rate scheduling, label smoothing, and exponential moving average weights to improve generalization and training stability. Model robustness is evaluated under realistic underwater and image-quality degradations, including Gaussian blur, JPEG compression, and synthetic turbidity, to assess the stability of predictions under challenging imaging conditions. Test-time augmentation and temperature scaling are further employed to improve prediction reliability and confidence calibration. A systematic ablation study is conducted to quantify the contribution of the proposed components relative to the ConvNeXt-Tiny baseline. CoralNet achieved a classification accuracy of 98.45%, demonstrating its effectiveness in distinguishing healthy and bleached coral images. The proposed framework provides an accurate and robust computer-vision-based approach for automated coral bleaching assessment, with potential applications in scalable coral reef monitoring and conservation.**

Keywords—Coral Reef, Coral Reef health Classification, Grad-CAM, Robustness, Deep learning, CNN, Explainable AI

I. INTRODUCTION

Coral reefs are among the most ecologically important marine ecosystems, supporting diverse marine communities and providing important ecosystem services, including fisheries resources and coastal protection. However, coral reef ecosystems are increasingly threatened by climate change, ocean warming, ocean acidification, declining water quality, and other local stressors. Elevated thermal stress can disrupt the symbiotic relationship between corals and their photosynthetic algae, resulting in coral bleaching and potentially severe degradation of reef ecosystems [1]. The increasing frequency and severity of environmental stressors therefore emphasize the need for effective and scalable approaches for monitoring coral condition.

Traditional coral-reef assessment frequently depends on manual analysis of underwater imagery by trained researchers. Although expert-based assessment provides valuable ecological information, large-scale manual annotation is labor-intensive and can limit the quantity and consistency of data that can be processed. Williams et al. demonstrated that automated image-analysis systems can substantially reduce the resources required for coral-reef monitoring while producing estimates that are highly comparable with human-generated assessments [2]. Similarly, the development of CoralNet demonstrated the feasibility of applying machine-learning and deep-learning approaches to large collections of underwater coral imagery. Chen et al. subsequently introduced a deep-learning engine for CoralNet based on transfer learning with EfficientNet-B0, demonstrating improved automated benthic-image classification compared with the previous system [3].

Deep learning, particularly convolutional neural networks (CNNs), has consequently become an important approach for automated analysis of underwater coral imagery. Previous studies have demonstrated the effectiveness of CNN-based models for coral recognition and classification. Raphael et al. applied deep neural networks to thousands of underwater images for automatic recognition of shallow-water coral species, demonstrating the potential of deep learning for large-scale coral monitoring [4]. Gómez-Ríos et al. investigated CNN architectures, transfer learning, and data augmentation for coral texture-image classification and demonstrated that deep convolutional models can achieve strong classification performance despite the limited size and complexity of available coral datasets [5]. More recently, deep-learning approaches have also been investigated specifically for distinguishing healthy and bleached coral images, highlighting the potential of automated image classification for coral-health monitoring [6].

Despite these advances, coral-image classification remains challenging because underwater imagery is strongly affected by environmental and imaging conditions. Variations in illumination, color distortion, turbidity, motion blur, image

resolution, and camera characteristics can substantially affect automated image analysis. The CoralNet deep-learning system, for example, identifies imaging conditions such as lighting, turbidity, motion blur, and camera quality as factors that can influence classification performance [3]. In addition, coral images exhibit considerable variation in morphology, texture, structure, and visual appearance, making it difficult for a model trained on one distribution to generalize reliably to different underwater conditions. Previous coral-classification research has also emphasized the difficulty associated with similar visual characteristics, inter-class variation, and differences between texture-dominated and structure-oriented coral imagery [5,7].

To address these challenges, this study develops a CNN-based framework for binary classification of healthy and bleached corals. Three publicly available coral-image datasets are integrated to construct a larger and more diverse dataset, followed by duplicate removal and underwater-oriented preprocessing. Data augmentation is subsequently employed during training to increase image variability and reduce the risk of overfitting. The use of multiple heterogeneous datasets is intended to provide a broader visual representation of healthy and bleached coral conditions and to improve the generalization of the resulting classification framework.

ConvNeXt-Tiny is adopted as the principal backbone of the proposed framework. ConvNeXt was introduced as a modernized pure convolutional architecture that incorporates design principles from contemporary visual-recognition research while retaining the simplicity and inductive biases of convolutional networks [8]. The original ConvNeXt study demonstrated competitive performance with contemporary transformer-based vision architectures while maintaining a purely convolutional design. This makes ConvNeXt-Tiny a suitable foundation for investigating modern CNN-based approaches to coral-image classification without introducing transformer architectures into the proposed framework.

The proposed framework further enhances convolutional feature extraction through attention-based and multi-scale feature representation. Coordinate Attention is incorporated to improve feature selection while preserving positional information, whereas a bidirectional feature pyramid structure is employed to facilitate the fusion of representations across multiple spatial scales. Such multi-scale representation is particularly relevant to coral imagery because bleaching patterns and coral structures can occur at different spatial resolutions. Global Average Pooling and Global Max Pooling are subsequently combined to capture complementary global and salient responses before classification. These architectural components are evaluated systematically rather than assuming that their individual inclusion necessarily improves performance.

The effectiveness of the proposed framework is evaluated through systematic ablation experiments in which the proposed components are progressively introduced to the ConvNeXt-Tiny baseline. This experimental design enables the contribution of attention, multi-scale feature fusion, training regularization, robustness evaluation, and inference strategies to be quantified independently. Through this framework, the study aims to develop an accurate and robust CNN-based approach for automated healthy-versus-bleached coral classification and to demonstrate the potential of modern convolutional computer vision methods for scalable coral-reef monitoring and conservation.

II. Related Works

Coral reefs are essential marine ecosystems hosting approximately 25% of all marine organisms, yet they are increasingly threatened by bleaching caused by rising water temperatures and environmental stressors [9], [10]. To combat this, researchers have utilized Convolutional Neural Networks (CNN) to classify corals into healthy, dead, and bleached states, emphasizing that the quantity of image data often outweighs quality in improving model performance [11]. For instance, a study focusing on the significant reef presence in the Philippines demonstrated that increasing the sample size and using balanced datasets boosted accuracy from 68.75% to 84.93%, utilizing architectures inspired by LeNet-5 that employ ReLU and max-pooling to extract prominent features [11].

Advancements in classification have led to hybrid frameworks that combine deep learning with handcrafted descriptors to better navigate the complexities of underwater imagery [10]. The RL-BoHVF (Resolute Localisation utilising Bag-of-Hybrid Visual Features) technique integrates the neural architecture of AlexNet with manually designed ColorTexture features using a Bag-of-Features (BoF) transformation [10]. This resolute localization method achieved a classification accuracy of 96.2% on Great Barrier Reef samples, providing researchers with a visual map of bleaching instances to help identify critical hotspots and measure the intensity of environmental damage [10].

Modern monitoring has shifted toward real-time object detection models like YOLOv8 and YOLOv9 to facilitate immediate ecological responses [9]. Research focused on the Red Sea applied these models to a highly augmented dataset of over 10,000 labeled images, where YOLOv9 achieved a slightly higher mean Average Precision (mAP) of 89% compared to YOLOv8 [Harnessing Computer]. A critical innovation in these systems is the ability to automatically notify authorities upon detecting distress, with future goals including integration with underwater drones for autonomous, high-speed reef assessment [9].

III. METHODOLOGY

Fig 1 illustrates the complete pipeline of the proposed model. We have used three openly available datasets and merged them to conduct this research. We have resized the images of the merged dataset, removed any duplicate data, and then split them into train, test, and validation sets. Then the images underwent preprocessing to improve feature extraction. We trained a lightweight model from the ConvNext family, with 28 million parameters, ConvNext-tiny [12].

After that, we applied the attention module CBAM so that the model dynamically focuses on important regions to make decisions and integrated GNN, allowing the model to do non-local pixel reasoning for prediction-making [13], [14]. Later, we tested the proposed model by applying Gaussian Blur, JPEG Compression, and Synthetic Turbidity on images to evaluate the robustness of the model. Finally, we generated Grad-CAM images for visual explanation by showing the highlighted area of the coral reef that influenced the model's prediction

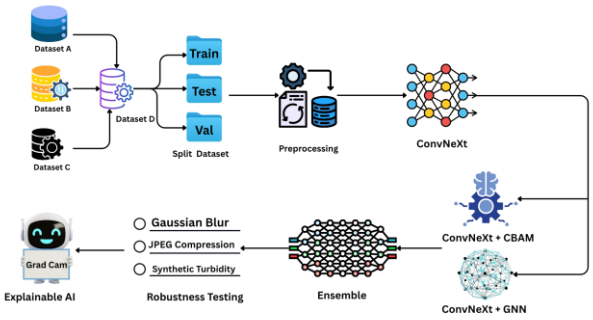


Fig. 1. Methodological framework

A. Dataset

In this study, we have used 3 datasets. Fig 2 shows a sample of Healthy and Bleached Corals Image Classification dataset containing a total of 923 images distributed into two classes: Healthy and Bleached [15]. The dataset is labeled as Dataset A.

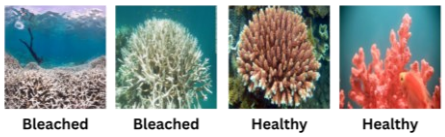


Fig. 2. Sample images from dataset A

Fig 3 illustrates a sample of the second dataset. Bleached and Unbleached Corals Classification dataset, labeled as Dataset B, contains two classes: the Bleached class with 184 images and the Unbleached class with 158 images [16].

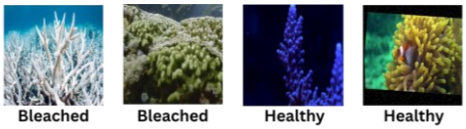


Fig. 3. Sample images from dataset B

The Coral Reef Bleach Detection Computer Vision Model dataset labeled as Dataset C contains a total of 10781 images distributed among Healthy and Bleached classes [17].



Fig. 4. Sample images from dataset C

All the images of the above dataset were taken in the underwater environment under various conditions and angles. All three datasets were merged into a single dataset so that the model can be trained on various data and can be more robust.

B. Preprocessing

All the images of the above dataset were taken in the underwater environment under various conditions and angles. All three datasets were merged into a single dataset so that the model can be trained on various data and can be more robust.

C. Model Architectures

ConvNeXt-tiny is the smallest variant of ConvNeXt, but the parameters of ConvNeXt-tiny are 28M; the network complexity of ConvNeXt-tiny is 4.5GFLOP[12].

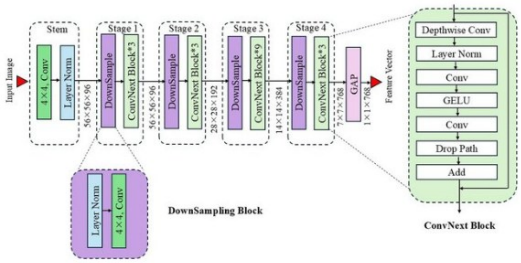


Fig. 5. ConvNeXt-tiny Architecture

It was selected as the backbone because it provides a modern convolutional architecture with strong visualization for moderate-sized datasets like our coral dataset. Fig 5 illustrates the architecture of the ConvNeXt-tiny model.

D. Attention Module

We have applied the CBAM attention module to extend our reliability. CBAM is a lightweight and general module that applies attention to the region of interest, so that the model focuses on the important areas to extract the necessary features [13].

E. Graph Neural Network (GNN)

Graph neural networks are deep neural networks designed for graphs with attributes attached to nodes or edges [14]. We have made a hybrid of CNN and GNN. As CNN extracts the features from the images, GNN takes them as nodes and maps edges between them and draws the relationships and dependencies among them.

F. Ensemble

We have used the Logistic Regression Stacking Ensemble to ensemble the ConvNeXt + CBAM model and ConvNeXt + GNN model [18]. Logistic Regression models the log-odds of a binary outcome as a linear combination, followed by a

sigmoid probability transformation. Equation 1 shows the logistic regression ensemble function used for this research :

$$P(Bleached\ |x) = \sigma(w0 + w1P1(\text{Bleached}\ |\ x) + w2P2(\text{Bleached}\ |\ x)) \quad (1)$$

where,

$$Class\ 1 = Bleached;\ Class\ 0 = Healthy.$$

Training Setup

We have created a new Dataset D by merging Dataset A, B, and C. It has increased the total number of images. The classes were equally balanced. The AdamW optimizer was used for all the models [19]. The batch size was set to 16.

TABLE I. HYPERPARAMETER SETTINGS FOR THE EVALUATED CNN

Architecture	Image Size	Batch Size	Loss Function (Focal Loss + Label Smoothing)		Optimizer	Early Stop (Patience)
			Focal Loss (γ)	Label Smoothing (∈)		
ConvNeXt	224	16	2.0	0.1	AdamW	15
ConvNeXt-Tiny + CBAM	224	16	2.0	0.1	AdamW	15
ConvNeXt-Tiny + GNN	224	16	2.0	0.1	AdamW	15

The Focal loss function was used as a loss function with label smoothing. A patience level of 15 was used for all models. Table I shows all the hyperparameters tuned for this research.

IV. RESULTS

In this section, we assess the models and techniques used in this research.

A. *ConvNext-Tiny*

Fig 6 illustrates the training and validation curve of the ConvNext-Tiny model.

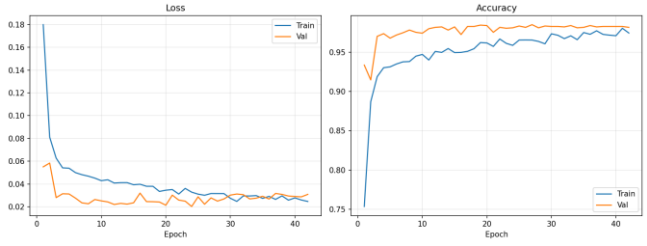


Fig. 6. Training and Validation Curve of ConvNext-Tiny

We can see that the model initially gained accuracy. The validation and training accuracy remained close, indicating that the model performs well with heavy augmentation, achieving 98.23% accuracy. The model's loss dropped a lot in the beginning and stabilized after 30 epochs.

Fig 7 shows the confusion matrix of the ConvNext-Tiny model [20]. Some misclassifications can be observed in both classes, but small mistakes are anticipated because of visual and environmental similarities among some coral reef images.

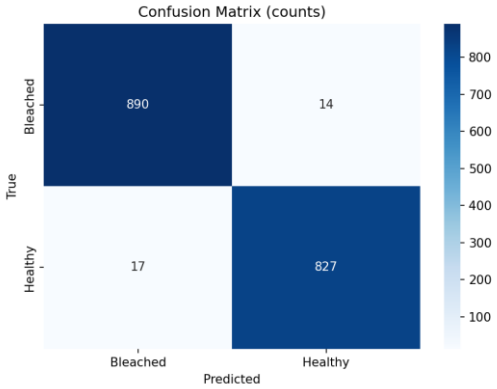


Fig. 7. Confusion Matrix of ConvNeXt-Tiny

B. *ConvNeXt-Tiny + CBAM*

Fig. 8 illustrates the training and validation curves for the ConvNeXt-Tiny + CBAM model.

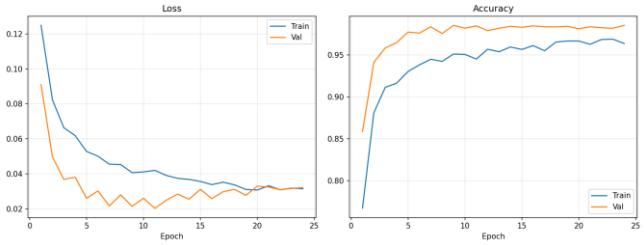


Fig. 8. Training and Validation Curve of ConvNeXt-Tiny + CBAM

The attention model helped the baseline model to extract the necessary features from the images, which is reflected in the accuracy curve, where the model gained accuracy rapidly as training progresses and stabilizes around 98.05%. The loss curve also shows the fast convergence of the model.

The confusion matrix of the ConvNeXt-Tiny + CBAM model is shown in Fig 9. This model also showed some misclassification, but these errors are anticipated due to the visual similarities on coral reefs in an underwater environment.

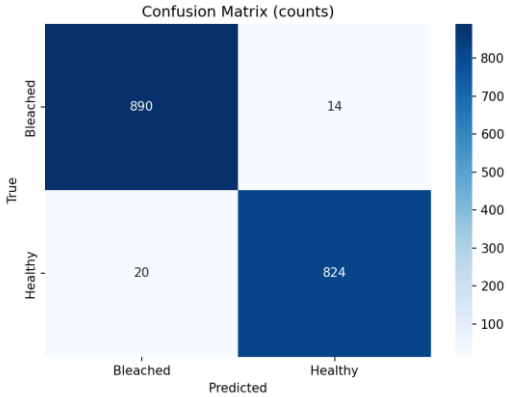


Fig. 9. Confusion Matrix of ConvNeXt-Tiny + CBAM

C. *ConvNeXt-Tiny + GNN*

The training and accuracy curve is shown in Fig 10, which shows a very promising result in terms of accuracy. We can observe that the validation and training accuracy curves converge very early and stabilize around 97.88% accuracy. Although this accuracy is lower than the previous one, it provides us with global features and their relationships. Then predicts based on the relationship. The loss curve tells us another story initially, the train and validation loss drop together, but after 22 epochs there is a gap between them. This is due to the hard augmentation.

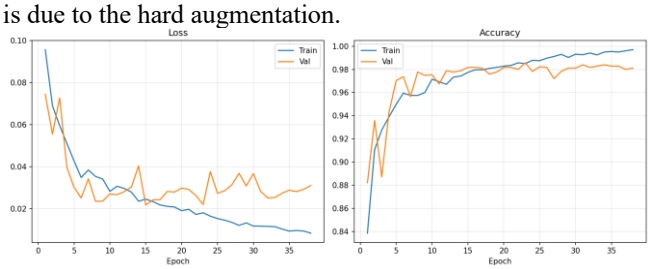


Fig. 10. Training and Validation Curve of ConvNeXt-Tiny + GNN

The attention model helped the baseline model extract the necessary features from the images, as reflected in the accuracy curve, which shows the model gaining accuracy rapidly as training progresses and stabilizing around 98.05%. The loss curve also shows the fast convergence of the model.

Fig 11 illustrates the confusion matrix of the ConvNeXt-Tiny + GNN model. In the confusion matrix, some errors can be observed, but these small errors can be foreseen due to the imaging similarities.

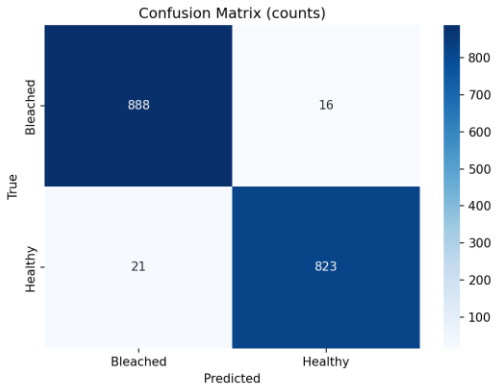


Fig. 11. Confusion Matrix of ConvNeXt-Tiny + GNN

D. *Proposed Model Performance*

To improve the performance and reliability, we have ensembled the ConvNeXt-Tiny + CBAM and ConvNeXt-Tiny + GNN models. Table II presents a comparative analysis between the models and the proposed model, CoralNet ((ConvNeXt-Tiny + CBAM) + (ConvNeXt-Tiny + GNN)), on the merged Dataset D. The baseline model ConvNeXt-Tiny's accuracy and macro-F1 score remained higher than those of the attention-integrated and GNN hybrid models. But after ensembling them, our proposed model CoralNet achieved the highest accuracy and macro-F1 score of 98.46% and 98.45%, respectively.

TABLE II. ADVERSARIAL ROBUSTNESS EVALUATION OF DENSENET201 UNDER VARYING PERTURBATION LEVELS (ε)

Model Name	Accuracy	Macro-F1Score
ConvNeXt-Tiny	98.23%	98.22%
ConvNeXt-Tiny + CBAM	98.05%	98.05%
ConvNeXt-Tiny + GNN	97.88%	97.88%
CoralNet	98.46%	98.45%

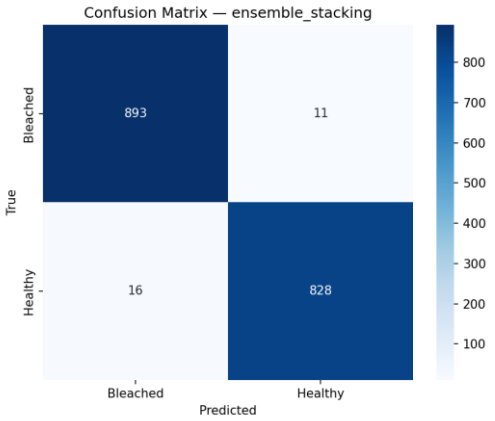


Fig. 12. Confusion Matrix of CoralNet

Fig 12 shows the confusion matrix of our proposed model, CoralNet. The model also showed some misclassification, but it is the lowest then the other models before. These few errors are expected due to environmental similarities in underwater situations.

V. ROBUSTNESS ANALYSIS

In this section, we will discuss our robustness analysis results under tests.

A. *Gaussian Blur*

When we added Gaussian blur to the images, the accuracy started to drop with different kernel sizes [21]. In Table III, we can see that as the kernel size increased, the accuracy decreased. The decrease rate was very mild, and our model was significantly steady, with the lowest accuracy of 96.84% at kernel size 11. The micro-F1 score remained the same as the corresponding accuracy.

TABLE III. ROBUSTNESS EVALUATION OF CORALNET ON GAUSSIAN BLUR

	Kernel Size	Accuracy	Macro-F1Score
Gaussian Blur	3	98.28%	98.28%
	5	98.34%	98.34%
	7	97.65%	97.65%
	9	97.30%	97.30%
	11	96.84%	96.84%

B. JPEG Compression

Table IV shows how JPEG compression affects the model accuracy with different quality levels [21]. We can see the model achieved 98.23% accuracy with the quality level of 80. As the image quality level decreased, the accuracy also decreased along with the macro-F1 score. The accuracy drop was not that rapid, but a massive drop can be observed at the quality level of 10, which is really promising for this type of low-quality image.

TABLE IV. ROBUSTNESS EVALUATION OF CORALNET ON JPEG COMPRESSION

	Quality	Accuracy	Macro-F1Score
JPEG Compression	80	98.23%	98.22%
	60	97.65%	97.65%
	40	97.37%	97.36%
	20	96.40%	96.39%
	10	90.10%	89.98%

C. Synthetic Turbidity

To test the model's reliability, Synthetic turbidity was introduced as a controlled domain-specific corruption to simulate reduced underwater visibility caused by suspended particles and light scattering [22]

TABLE V. ROBUSTNESS EVALUATION OF CORALNET ON C. SYNTHETIC TURBIDITY

	Strength	Accuracy	Macro-F1Score
Synthetic Turbidity	0.1	98.46%	98.45%
	0.2	98.23%	98.22%
	0.3	98.23%	98.22%
	0.4	98.00%	97.99%
	0.5	97.88%	97.88%

In Table V, we can see the model has shown a very steady performance. As the strength level increased, the model maintained a consistent accuracy.

At strength level 0.5, the model gained 97.88% accuracy, which illustrates how reliable the model is.

VI. EXPLAINABLE AI

In addition to increasing the reliability and explainability of the model, we used a method called Grad Cam so that the model doesn't act like a black box [23]. This helps us to determine which part of the coral affects the prediction.

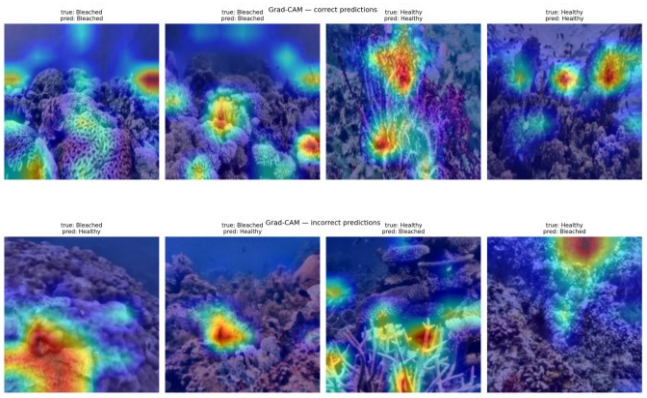


Fig. 13. Coral Health condition identification using Grad Cam

VII. COMPARISON AND DISCUSSION

In this section, we have done a comparative analysis of our proposed architecture with previous works on coral reef health classification on different architectures. This will help us to assess the effectiveness of our model.

A. Compare Against Prior Work

A benchmark analysis has been done on CoralNet and prior research works. CoralNet achieved 98.45% accuracy out performing many previous methodologies shown in Table VI For instance, Jon Borbon et al. [11] achieved 84.93% accuracy on a CNN-based architecture. Another researcher, Afnan Aldhahri et al.'s [9] YOLOv8 model achieved 89% accuracy. Subash Banala, et al. [10] reached an accuracy of 96.20% with his method RL-BoHVF. CoralNet demonstrated high accuracy while maintaining transparency.

TABLE VI. COMPARISON OF CORALNET WITH PREVIOUS WORKS ON CORAL HEALTH CLASSIFICATION

Author	Method	XAI	Accuracy
Afnan Aldhahri, et al. [9]	YOLOv8	No	89%
Subash Banala, et al. [10]	RL-BoHVF	No	96.20%
Jon Borbon, et al. [11]	CNN	No	84.93%
Proposed Model	ConvNext-CBAM + ConvNext-GNN	Yes	98.45%

B. Discussion of Limitations and Future Work

This study merges three coral datasets with different cameras and conditions, risking domain shift that could not be fully verified due to lost source tracking after merging. Mixed cropped and full-scene images were handled using attention rather than explicit region detection, and binary labeling ignores partial bleaching severity, which is common in real reefs. Evaluation relied on a single held-out split without independent out-of-distribution testing, and the CNN+GNN component adds computational overhead unsuitable for real-time deployment.

Future work should preserve source labels for cross-dataset validation, add region-based coral detection, extend to graded bleaching-severity estimation, incorporate environmental and temporal data, and validate attention-based interpretability

against marine-biology expert judgment before field deployment.

VIII. Conclusion

This study presented a comprehensive deep learning pipeline for coral bleaching classification, combining underwater-specific preprocessing, attention mechanisms, and architectural diversity across three merged datasets. Among individual backbones, ConvNeXt-Tiny achieved the strongest standalone performance (98.23% accuracy, 98.22% F1), while CBAM and GNN variants offered complementary relational and spatial reasoning despite marginally lower individual scores (98.05% and 97.88% respectively). Ensembling these architectures into CoralNet yielded the best overall result (98.46% accuracy, 98.45% F1), confirming that architectural diversity outperforms any single model. These findings demonstrate that combining attention-based and graph-based reasoning with robust preprocessing yields a reliable, generalizable framework for automated coral health assessment, supporting scalable reef-monitoring efforts.

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